Introduction to Ergonomics and Human Factors

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Ergonomics is an applied scientific discipline that is concerned with how humans interact with the tools and equipment they use while performing tasks and other activities.[1] A common focus of ergonomics is the interface between humans and equipment—for example, the controls, displays, and other devices that are used by workers to operate the equipment. Our primary interest is on the activities that humans perform while working. These work activities impose a continuum of physical and cognitive demands on the human worker, as represented by the scale in Figure 1. In fact, most human activities require a combination of physical and cognitive exertions. Ergonomics is also concerned with the physical and social environment in which the tasks and activities are performed and how humans and machines interact with the environment. The word is derived from two Greek words, *ergon*, meaning work, and *nomos*, meaning rules or laws.[2] Combined, the words mean (roughly) the study of work. A professional whose practice is primarily focused on ergonomics is called an ***ergonomist.***

A term frequently used in place of ergonomics is ***human factors.*** Until recently, this term was most commonly used in the United States, while "ergonomics" was the

[1]The Work Measurement and Methods Standards Subcommittee (ANSI Standard Z94.12-1989) defines ergonomics as follows: "The application of a body of knowledge (life sciences, physical science, engineering, etc.) dealing with the interactions between man and the total working environment, such as atmosphere, heat, light and sound, as well as all tools and equipment of the workplace."

[2]The word ***ergonomics*** entered the English language in 1949. It was coined by British scientist K. F. H. Murrell, although it had been used first by Polish scientist W. Jastrzebowski in an 1857 newspaper article.

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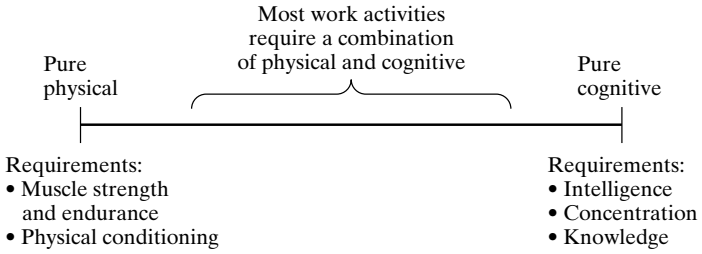


Figure 1 Continuum of physical and cognitive demands that work activities impose on humans. Based on a figure in [13].

preferred term in Europe.[3] Today it is generally accepted that ergonomics and human factors are synonymous (Historical Note 1). If there was a difference in these terms, the difference was that ergonomics emphasized work physiology and anthropometry,[4] while human factors emphasized experimental psychology and systems engineering [7]. Also, the interest and applications in European ergonomics was more on industrial work systems, while the interest and applications in human factors in the United States was more on military systems.

HISTORICAL NOTE 1 HISTORY OF ERGONOMICS

Ergonomics as an identified field of study can be traced back to around 1945, although some of the motivating influences precede this date. One of the most important was no doubt the scientific management movement, or ***Taylorism***,[5] with its emphasis on task planning, motion and time study, and worker efficiency. Critics have argued that Taylorism created industrial jobs that were simplistic and repetitive, and that it dehumanized labor. On the other hand, it made possible an industrial system in the United States that was the most productive in the world. And it focused attention on work as an area of study and research. Included in the scientific management movement were the pioneering studies of Frank and Lillian Gilbreth in the area of motion study, which has been referred to as "one of the forerunners" of human factors [15].

In the early 1900s, there was considerable effort in industry devoted to the selection of workers for a given job. This was called "fitting the man to the job" (FMJ). The idea was to choose from the pool of job applicants those individuals who were best suited to the requirements of the position on the basis of their physical and/or mental

[3]It is of interest to note that the Human Factors Society in the United States changed its name in 1992 to include the word ergonomics. The society is now called the Human Factors and Ergonomics Society.
[4]These terms are defined in Section 3.
[5]Named after Frederick W. Taylor, the father of scientific management.

aptitudes. Psychological testing was often used in the selection process when intelligence and personality were among the job requirements.

Studies carried out in the late 1920s at the Hawthorne Works of Western Electric Company near Chicago drew attention to the importance of social factors in the workplace. The Hawthorne experiments as they came to be known represented the beginning of interest in "human relations" research.

Finally, another important influence leading up to ergonomics was the growth in the use of machinery and mechanization between 1900 and 1945. The automobile industry was the significant growth industry at the beginning of this period, and the production of cars was enabled by mechanized machine tools and conveyorized assembly lines. In the 1930s, World War II forced nations to develop modern production technologies to meet the demand for munitions. Although the United States did not officially enter the war until the end of 1941, it was supplying weapons to its future allies during the 1930s.

Recognition of an emerging discipline related to humans working with machines crystallized around the end of World War II. It was called *ergonomics* in Europe and *human factors* in the United States. The war had witnessed a dramatic increase in the technological complexity and sophistication of weapons systems, such as aircraft, artillery, tanks, radar, and sonar. Significant problems were sometimes encountered in the operation of this equipment by humans, with loss of life too often the result. These problems only increased with the new jet aircraft that were being developed after the war. There was a need to deal with a new problem area known as *human–machine systems*. The interest in the United States was on aircraft and other weapons systems, and human factors research was frequently associated with "knobs and dials" between 1945 and 1960. The Human Factors Society was established in the United States in 1957.

Meanwhile, the field of ergonomics was starting in Britain. K. F. H. Murrell, an early contributor in the field, is credited with giving the discipline its name in 1949. The emphasis of ergonomics in Europe was on industrial work systems, including equipment and workspace design used in such systems. In 1950, British researchers met to discuss a new professional society to represent the discipline, and the name Ergonomics Research Society was adopted in that year (the name was later shortened to Ergonomics Society).

By the 1960s, the space age had arrived and the computer industry had begun its growth spurt. In the United States, human factors started to be applied outside of the military. The National Aeronautics and Space Administration (NASA) had been established in 1958 for the exploration of space, and human factors were quickly identified as a concern in manned space flight. Industrial applications of human factors also grew during the decades following 1960, to design both work systems and consumer products. A human factors group was established in the early 1960s at Eastman Kodak Company, the first company to form such a department.

Between 1980 and the present, interest in ergonomics and human factors has continued to grow, motivated on the one hand by advances in computer and automation technologies and on the other hand by several disasters that have highlighted the critical importance of the human in the operation of advanced human–machine systems.

In a computer system, the computer is the machine with which the human interacts. Human factors research has focused on the human-computer interface and the effective exchange and processing of information between these components. Several textbooks have been written on this subject, including two that are listed among our references [4, 6].

Since 1979, the following significant human-related disasters have contributed to public awareness about the importance of ergonomics and human factors:

- Three Mile Island nuclear power plant accident (near Harrisburg, Pennsylvania, March 1979)
- Bhopal pesticide plant, Union Carbide Company (Bhopal, India, December 1984)
- *Challenger* space shuttle (above Cape Kennedy, Florida, January 1986)
- Chernobyl nuclear power plant explosion (Kiev, Ukraine, April 1986)
- *Exxon Valdez* oil spill (Prince William Sound, Alaska, March 1989)
- *Columbia* space shuttle (over Texas, February 2003).

Many of these disasters were the result of human errors, sometimes caused by inadequate attention to ergonomics and human factors in system design and system management, and other times caused by flaws in decisions by humans.

This chapter provides an overview of the field of ergonomics and human factors, including its history and main topical areas. It then focuses on the human–machine system, an important model in the application of ergonomics.

1 OVERVIEW OF ERGONOMICS

In this section, we discuss the objectives and applications of ergonomics, and how the field has evolved into an important area in work design that attempts to adapt the task to the worker's capabilities.

1.1 Objectives and Applications of Ergonomics

The main objective of ergonomics is to improve the performance of systems consisting of people and equipment. Such systems are often referred to as human–machine systems (Section 2), although the word *machine* includes a variety of objects that are not normally thought of as machines—for example, aircraft, appliances, automobiles, chairs, computers, hand tools, and sports equipment. Better performance of human–machine systems in an ergonomics context means the following:

- Greater ease of interaction between the user and the equipment
- Avoidance of errors and mistakes by the user
- Greater comfort and satisfaction in the use of the equipment

- Reduced stress and fatigue while using the equipment
- Greater efficiency and productivity
- Safer operation of the equipment
- Avoidance of accidents and injuries.

These objectives are similar to those in methods engineering. Methods engineering and ergonomics are closely related and their general objectives are the same: (1) to improve the performance of existing systems and (2) to design new systems for optimum performance. Whereas methods engineering emphasizes issues such as efficiency, cost reduction, and elimination of waste, ergonomics emphasizes the human aspects such as comfort and safety and the interactions that occur between the human, the equipment, and the environment.

Ergonomics has a wide variety of applications, nearly all of which can be classified into two main areas:

- *Work system design.* This area is concerned with the interaction between worker and the equipment used in the workplace. Objectives include safety, accident avoidance, and related performance attributes. Work system design includes consideration of factors related to the work environment such as lighting and noise levels.
- *Product design.* This area deals with the design of products that are safer, more comfortable, and more user-friendly and mistake-proof. In addition to providing greater customer satisfaction by means of these kinds of features, an issue in product design is product liability lawsuits and their avoidance through consideration of ergonomics.

In our treatment of ergonomics here, the focus is on work systems, which in fact overlap with product design in the sense that work systems usually include equipment, and equipment is designed and fabricated by companies that are selling it as a product. A production machine is a product that must be designed with ergonomics in mind if it is to operate productively and safely.

1.2 Fitting the Job to the Person

A common employment practice prior to ergonomics was based on a philosophy called "fitting the person to the job" (FPJ), which recommended that workers be selected on the basis of their mental aptitudes and physical characteristics for a particular job opening.[6] Psychometric testing (e.g., tests for intelligence and personality characteristics) had begun to be used following World War I in the hiring process to improve productivity by trying to identify workers who had the right psychometric capabilities for the particular job. A worker's physical attributes were also used in the selection process for jobs requiring characteristics such as size and strength. The FPJ approach is still considered among the eligibility factors for certain positions in many hiring situations today; for example, firefighters must meet physical endurance requirements and

[6]The original term used was "fitting the man to the job" (FMJ). To observe modern gender-neutral language conventions, we have changed the name to "fitting the person to the job" (FPJ).

candidates seeking to become military pilots must meet height requirements. Educational qualifications are also widely used in selecting employees for certain positions such as engineering and public school teaching, and professional certification is required for many occupations, including medical doctors, dentists, and lawyers. All of these examples represent situations in which mental, physical, and/or educational requisites are used to ensure that the person is qualified for a given job.

The ergonomics approach is diametrically opposite of FPJ. The philosophy in ergonomics is "fitting the job to the person" (FJP)—that is, designing the job so that nearly any member of the workforce can perform it. There are several factors that explain why the new philosophy has evolved and now occupies a position that operates in parallel with and sometimes supersedes the FPJ approach: (1) changes in worker skill requirements, (2) demographic changes, and (3) social and political changes.

In the early decades of the 1900s, the general skill requirements for many jobs were much lower than today and there were many applicants, including many young people, for those low-skilled positions. Companies could be much more selective when hiring their workers. Today, the skill requirements of employment are much higher. Whereas the work in earlier times was largely physical and clerical, today it involves the operation of machinery and computers. The equipment is often sophisticated, and the work processes are complex. The average age of the available workforce is older than in the early 1900s. Companies are often faced with a smaller pool of applicants to choose from, and they cannot afford to be as choosy. They must hire from the pool those who are qualified but not necessarily optimal in the FPJ sense. Training must be provided, and the work itself must be designed to be readily understood so that the necessary amount of training can be minimized and workers can be productive as soon as possible after employment begins.

One significant manifestation of the demographic changes in employment has been the increase in the number of working women in the latter half of the twentieth century. More women have been entering the workforce in positions heretofore dominated by male workers. Women have been hired as assembly line workers and machine operators, and they have attended college in growing numbers to occupy positions as engineers and managers. Table 1 provides some statistical data to show the demographic changes that have taken place between 1930 and 2000.

At the same time that these changes in skill requirements and demographics were occurring, social and political pressures were effecting changes in cultural attitudes and government regulations. Equal opportunities legislation was enacted to prevent discrimination on the basis of race, gender, and age. Laws were enacted at the state and

TABLE 1 Changes in U.S. Population and Workforce Between 1930 and 2000

	1930	2000
Total population in the United States	123 million	281 million
Life expectancy	60 years	77 years
Median age	27 years	35 years
Number of people aged 65 and over	7 million	35 million
Proportion of women in the labor force	22%	61%

Source: U.S. Census Bureau. (www.census.gov/pubinfo/www/1930_factsheet.html, March 28, 2002.)

federal levels to mandate that physically handicapped persons be provided with access ramps and other conveniences at public facilities and in transportation. Hiring of the handicapped was encouraged. In some cases, organizations went beyond the legislation by pursuing proactive policies on the hiring of minorities and the handicapped.

2 HUMAN–MACHINE SYSTEMS

Worker–machine systems, from an operational perspective, can focus on issues such as the kind of work done and the time factors involved. In this section, we consider the same basic work systems but from an ergonomics viewpoint. The basic model in ergonomics is the ***human–machine system***, defined as a combination of humans and equipment interacting to achieve some desired result. The number of humans can range from one to many, and the type and amount of equipment can range from a single hand tool to a complex and sophisticated collection of machines and computers. The desired result achieved by the system refers to its purpose and/or function—for example, to produce a product or provide a service. A human–machine system can have multiple objectives—for example, to produce a product, to contribute to the profit of the organization, and to provide a safe and comfortable workplace.

Human–machine systems can be classified into three basic categories [15]:

1. *Manual systems.* This system involves a person using some hand tool or other nonpowered implement to perform an activity. An example is a farmer using a pitchfork to load hay into a wagon.
2. *Mechanical systems.* This system refers to one or more humans using powered equipment to accomplish some job. The typical case is that the equipment provides the mechanical power for the job, and the function of the human(s) is to control the equipment. An example is a farmer driving a tractor to harvest a crop.
3. *Automated systems.* This system involves the performance of a job with a minimum of human attention. The truth is that automated systems do require occasional human attention, for purposes of maintenance and repair, reprogramming, or loading and unloading of parts from a production machine. In addition, humans are required to design and install automated systems. Thus, even automated systems are human–machine systems.

A key feature of a human–machine system is that interactions occur between the humans and machines, as depicted in Figure 2. The human controls the machine operation, and the machine displays its actions. The display may consist of the operator simply observing the operation, or it may involve some sort of display device such as a television monitor. Whatever the case, the human senses the operation, processes the sensed information, and performs actions on the machine controls that regulate its operation. The interactions consist of a cycle of activities accomplished by the human and the machine. The interactions occur in an environment—both physical and social—that may influence system performance. For example, poor lighting may impede a worker's ability to perform an inspection task; an unfriendly supervisor may reduce a worker's motivation to work.

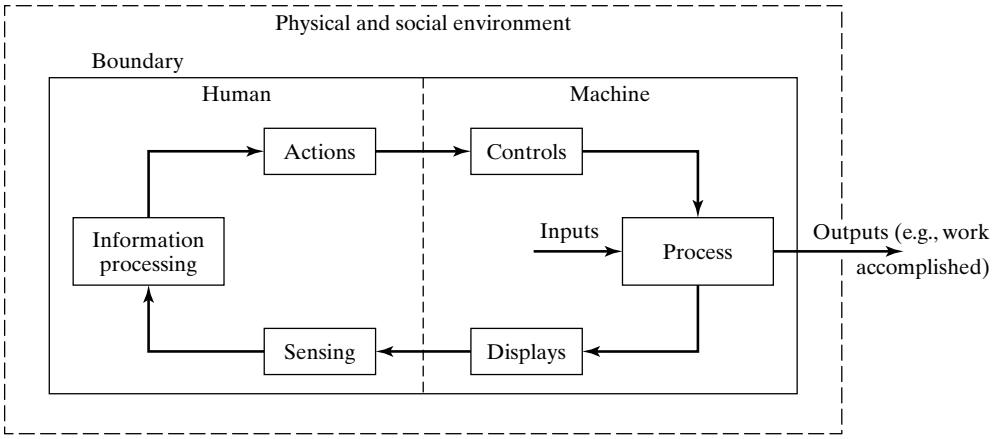


Figure 2 Block diagram model of the interactions in a human–machine system.

From an ergonomics design viewpoint, a human–machine system has boundaries, as shown in Figure 2, that define what components are included within the scope of the system for purposes of analysis and design. Most systems are components of a larger system. There is a systems hierarchy. A worker–machine production cell is one component in the larger production department, which is only one of many departments in the company. The ergonomist must decide where to draw the boundaries of the human– machine system of interest. The boundaries separate the human–machine system from its environment.

The two main components of a human–machine system are the human and the equipment. In turn, the human and the machine have components that must be considered in the context of the interactions occurring between them. The third component is the environment, which interacts with the human–machine system. The ergonomic objective is to optimize the performance of the system by making the interactions between the human and the equipment as seamless as possible while the system operates in its environment.

Human Components. As suggested by the model of the human–machine system in Figure 2, the human components of interest are those that perform three functions: (1) sensing the operation, (2) information processing, and (3) actions. The corresponding human components are the following:

1. *Human senses.* The five basic human senses are vision, hearing, touch, taste, and smell. They are the sensors by which humans are able to be aware of where they are and what is happening around them. Vision and hearing are the most important in ergonomics, although the others are sometimes useful—for example, sensing that a machine is running too hot or that it is emitting noxious fumes. People use a combination of their senses in many of their activities. For example, when we eat a meal at a fine restaurant, we enjoy not only its taste but also the visual presentation of the food and its aroma.

2. *Human brain.* The human brain can be considered as a data- and information-processing unit, analogous in some ways to the operation of a computer. It receives data through the human senses as well as feedback from the joints and muscles of the body (data entry). It stores the data in memory (computer storage). It operates on the data (as in the operation of a computer's central processing unit) by thinking, planning, calculating, solving problems, and making decisions about what actions to take. These thought processes include both reflex and cognitive activities (low-level and high-level software). The decisions are communicated to and executed by the human effectors (computer peripherals).
3. *Human effectors.* An effector is a body part such as a muscle or group of muscles that actuates in response to some stimulus. The principal human effectors in the human–machine system are the fingers, hands, feet, and voice. These are supported by the musculoskeletal system of the body, and the stimulus is provided by the information processing occurring in the human brain.

These human components operate in a coordinated fashion and interact with the machine components in the human–machine system.

Machine Components. As indicated previously, the machine in the human–machine system can range from a simple hand tool to a complex and sophisticated system of equipment. The typical model in ergonomics is one in which the interactions between the human and the machine are directly coupled. The following are three common examples:

- A worker using a shovel to dig a hole in the ground. The digger maintains continuous control over the shovel, using arm and body strength to manipulate it and accomplish the hole-digging activity.
- A person driving a car. The driver continuously steers the car using the steering wheel and controls its speed using the throttle pedal, and the car provides feedback such as direction, speed data, and engine rpm. The connections between the car and driver are tangible as the car moves down the road.
- A student writing a term paper on a PC. The student types in the text using the alphanumeric keyboard, and the computer responds by displaying the text on the monitor and identifying spelling and grammar errors.

In other cases, the coupling is less direct and the interactions are less tangible, as in the following two examples:

- A worker monitoring the operation of an automated chemical process on a computer terminal. The automated process proceeds without human intervention. The worker's function is to make sure that each process variable remains within a defined tolerance of its target value and to take some action (directly interact with the process) only if the tolerance is exceeded or some other exception occurs.
- A researcher using the Internet to search for articles on a topic of interest. The researcher enters keywords about the topic, and a search engine looks on the

Internet to find Web sites that provide links with the keywords. The search engine is guided by its own software, independent of any further input from the researcher.

As Figure 2 indicates, there are three principal components of the machine in the human–machine system:

1. *The process.* The process refers to the function or operation performed by the human–machine system, such as digging a hole, driving a car, writing a term paper, performing a chemical process automatically, or searching the Internet.
2. *Displays.* For processes that are simple and the worker participates directly, the display function is accomplished by the worker observing the process as it is being performed. No additional display of information is needed. For example, the worker digging a hole sees the depth and shape of the hole as each shovelful is removed and proceeds with digging until the proper hole size has been achieved. For processes that are more mechanized and automated, there is a greater need for artificial displays to allow the human to see and understand what is going on. Driving a car requires a combination of natural and artificial displays. The driver receives input about the process directly by observing how the car is tracking the road ahead and indirectly by observing the speedometer and other dashboard gauges. Highly automated processes like the chemical process in our example require the worker to obtain the necessary information entirely from the display monitor because the process itself cannot be directly observed.
3. *Controls.* The human interacts with the equipment using machine controls that can be operated by the human effectors, usually the fingers and hands. When the equipment is simple (e.g., a hand tool), the controls often consist of one of the equipment components. In our example of a worker digging a hole, the handle of the shovel is the only control that is manipulated by the worker. As the machine becomes more complex, the controls tend to be more remote from the actual operation performed by the human–machine system. Examples of these remote controls include the steering wheel of an automobile, the alphanumeric keyboard of a personal computer, the foot pedals of a church organ, and the levers on an engine lathe for feed and speed.

Environmental Components. A human–machine system operates in an environment that includes two components:

- *Physical environment.* The physical environment includes the immediate area of the human–machine system, separated from the system by a defined boundary in Figure 2. Components of the physical environment usually include the location and surrounding lighting, noise, temperature, and humidity. These environmental factors can affect the performance of the human–machine system and are of interest to the ergonomist. For example, the workspace of a fighter aircraft is its cockpit, which imposes severe limitations on the freedom of movement of the

pilot. The many controls of the aircraft must be located within easy reach of the pilot.

- *Social environment.* Work serves a social function. It is a chance for workers to meet and interact with other people. The primary work interactions are job-related, but social interactions occur also. The social environment at work is determined by coworkers and colleagues, immediate supervisors, organizational culture, and the work organization. The work organization consists of factors such as whether workers work alone or in teams, the required pace of the work, and whether the pace is set by the worker or by a machine.

3 TOPIC AREAS IN ERGONOMICS

As an academic subject, ergonomics has a breadth and depth that are much greater than we can do justice to in the space we have allocated to the subject in this text. For the interested reader, there are a number of good textbooks listed among our references as well as a handbook devoted to the subject. In our coverage, we emphasize the following topics.

Physical Ergonomics. Many work activities require physical exertion. The term *manual labor* is often used to refer to these activities. ***Physical ergonomics*** is mostly about manual labor. It is concerned with (1) how the human body functions during physical exertion and (2) how the physical dimensions of the body affect the capabilities of the worker. These two areas are related to the sciences of human physiology and anthropometry.

Physiology is the branch of biology that deals with the vital processes carried out by living organisms and how their constituent tissues and cells function. Our interest in physiology is directed at how the human muscles function during work activities. There are strength and endurance limitations of the muscles that must be considered in the design of manual work. The energy expenditure that is required in the activity must be in balance with the capacity of the body's cardiovascular and respiratory systems. Rest periods must be allowed for work periods involving strenuous physical output. There are guidelines and formulas that can be used to determine the appropriate rest periods for work of a given physical energy expenditure.

Anthropometry is the branch of anthropology that is concerned with the physical dimensions and other data related to the human body, such as height, reach, and weight. The data for a given dimension is normally distributed for a defined population (e.g., heights of females living in Northern Europe), but there are differences in these dimensions between defined populations (e.g., males versus females living in Northern Europe). There are several ergonomic design principles that relate to anthropometric data. One widely applied principle is "design for extreme individuals," which recommends that the designed entity should accommodate virtually all of the user population. For example, doorways should be designed for very tall people, even though the vast majority of users who pass through them are not very tall.

Cognitive Ergonomics. Most work activities also include cognitive elements as well as physical elements. Even manual labor usually requires some cognitive effort. Human activities with a high cognitive component include thinking, reading, speaking, learning, problem solving, and decision making. ***Cognitive ergonomics*** is concerned with the capabilities and limitations of the human brain and sensory system while performing activities that include a significant amount of information processing. It is the ergonomics area that considers how the human brain receives information from the environment, processes that information, and determines appropriate responses and actions. The human cognitive processes include the following components:

- *Sensory system.* This system consists of the five senses—vision, hearing, touch, smell, and taste—that are activated by external stimuli, such as sights and sounds. The vast majority of human information input is by sight and hearing.
- *Perception.* This is the stage of cognition that follows the sensing of external stimuli. It occurs when the human mind becomes aware of the sensation and interprets it based on previous experience and knowledge.
- *Memory.* Human memory consists of two main components: working memory and long-term memory. Working memory is temporary memory that holds a limited amount of information while being processed. It transfers information to and retrieves information from long-term memory, which is the warehouse of one's knowledge and experiences.
- *Response selection and execution.* This is the cognitive process of figuring out what actions are needed in the light of information perceived through the sensory system and processed through working memory and long-term memory.

The functions and operations of these components are not described completely here, nor are the number of ergonomic design guidelines that are derived from an understanding of them.

The Physical Work Environment. As noted in Figure 2, work is performed in an environment that includes both physical and social aspects. The physical work environment can be classified into the following components: (1) the visual environment, (2) the auditory environment, and (3) the climate.

The ***visual environment*** is what the worker sees while working. Most of the information input perceived by the worker is through visual stimuli. The important factor enabling the proper functioning of a worker's sense of vision is lighting. The term that describes how well things can be seen is ***visibility***. The visibility of tools, materials, and other items in the workplace depends on lighting levels, brightness contrast, color, and glare. The job of the ergonomist is to determine the optimum illumination levels for the task and to implement other design details in the workplace (e.g., types of lamps, colors, glare reduction) in order to achieve the best possible visual environment.

The ***auditory environment*** is what the worker hears while working. Much of the information input received by the worker is through the sense of hearing. Auditory information input is the second most important after visual in most work situations. Included

in the auditory environment is ***noise***, which can be defined as sound that is undesired and possibly harmful to a worker's sense of hearing. The possible effects of noise range from minor distraction to major deafness. The most important factors about noise in terms of its effects on humans are (1) intensity, perceived as loudness, and (2) duration of exposure. The Occupational Safety and Health Administration has established permissible noise levels that define the duration of exposures allowed for various intensity levels. Employers must make provisions to control noise in the workplace through various administrative and engineering approaches.

The ***climate*** is what the worker feels while working. Climate in the workplace environment is defined by four primary variables: (1) air temperature, (2) humidity, (3) air movement, and (4) radiation from surrounding objects (e.g., machines that operate at elevated temperatures and radiate heat). To a limited extent, the human body can compensate for variations in the normal body temperature of 35°C (98.6°F) through such mechanisms as perspiration when body temperature is above normal and shivering when body temperature is below normal. However, the climatic conditions that cause heat stress and cold stress often require interventions beyond the body's natural responses. The interventions include proper clothing for the conditions, frequent rest breaks, and heating and air conditioning of the facility.

Occupational Safety and Health. Every year in the United States, approximately 5000 lives are lost due to occupational injuries, and about 50,000 lives are lost due to work-related diseases and illnesses. Occupational safety and health constitute an important national issue that affects virtually every person who works. Several government agencies have been established to address the issue. The following federal agencies were established with the enactment of the Occupational Safety and Health Act in 1971:

- Occupational Safety and Health Administration (OSHA), whose responsibilities include enforcing the provisions of OSHAct, implementing safety and health programs, and setting occupational safety and health standards.
- National Institute for Occupational Safety and Health (NIOSH), whose responsibility is to engage in research, training, and education in the area of occupational safety and health.

The topics of occupational safety and occupational health are certainly related, but distinctions can be made between them. ***Occupational safety*** is concerned with the avoidance of industrial accidents, which are one-time events that can cause injury or fatality. The occupational safety objective in ergonomics is to understand how and why accidents, injuries, and fatalities occur and to take the necessary steps to prevent them. ***Occupational health*** is concerned with avoiding diseases and disorders that are caused by workers being exposed to hazardous materials or conditions in the workplace. These diseases and disorders do not generally occur immediately upon exposure. Instead they tend to develop after prolonged periods of exposure to the materials or conditions. The effects of the exposure are cumulative and may take years before symptoms reveal the onset of the malady.

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REVIEW QUESTIONS

1 Define the word *ergonomics*.

2 What are some of the objectives and benefits of ergonomics as applied to human–machine systems?

3 Ergonomics has two main application areas. Identify and briefly describe them.

4 What is meant by the work philosophy "fitting the person to the job?"

5 What is a human–machine system?

6 What are the three basic categories of human–machine systems?

7 What are the basic functions performed by humans in the human–machine system?

8 What are the three components of the human that correspond to the three functions?

9 What are the principal components of the machine in the human–machine system, as indicated in the diagram of the system?

10 Briefly describe the physical environment and the social environment in which the human–machine system operates.

11 Define physical ergonomics as this term is used in the text.

12 What is anthropometry?

13 Define cognitive ergonomics as the term is used in the text.

14 What are the three main components of the physical work environment?

15 What is the difference between occupational safety and occupational health, as these terms are distinguished in the text?

Occupational Safety and Health

1 Industrial Accidents and Injuries
  1.1 Human Error and Accidents
  1.2 Job Factors
  1.3 Environmental Conditions
2 Occupational Disorders and Diseases
3 Occupational Safety and Health Laws and Agencies
  3.1 Workers' Compensation Laws
  3.2 Occupational Safety and Health Act
4 Safety and Health Performance Metrics

One of the important goals of ergonomics and human factors is to provide a safe workplace. Safety and a healthful working environment are not only desirable social objectives; they are also a significant cost issue for industry and government. Consider the following:[1]

- Each day, an average of 16 workers die from injuries that occurred while working. This amounts to approximately 5000 lives lost per year due to occupational injuries.
- These deaths and injuries are estimated to cost society more than $100 billion per year, not to mention the personal upheavals and tragedies inflicted upon the families involved.
- Each day, an average of 9000 workers in the United States suffer injuries on the job. This amounts to nearly 5 million workers per year.
- The number of workdays lost due to occupational injuries is between 50 million and 100 million per year.
- Each day, an average of 137 workers die from work-related diseases. This adds up to approximately 50,000 lives lost per year due to occupational illnesses.

[1]Compiled from Web sites for Bureau of Labor Statistics (www.bls.gov), Occupational Safety and Health Administration (www.osha.gov), and National Institute for Occupational Safety and Health (www.cdc.gov/niosh), based on 2002 data.

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Although occupational safety and occupational health are related topics, they should be distinguished. ***Occupational safety*** is concerned with the avoidance of industrial accidents and in particular accidents that cause injury or fatality. These accidents are one-time events. The goal in ergonomics is to understand how and why they occur and to undertake steps to prevent them. ***Occupational health*** is concerned with avoiding diseases and disorders that are induced by exposures to materials or conditions in the workplace. These sicknesses and ailments do not generally occur immediately but instead develop after prolonged periods of exposure to the materials or conditions. The effects of the exposure are cumulative.

Our chapter begins with a discussion of industrial accidents and injuries: what causes them, and what steps can be taken to reduce their frequency and severity. The second section deals with disorders and diseases that result from exposures in the workplace. Next we review the important laws that have been enacted in the area of industrial safety and health. The federal agency that is most directly concerned with industrial safety and health is the Occupational Safety and Health Administration (OSHA). Finally, we define some of the measurements that are used to compile statistics about occupational safety and health.

1 INDUSTRIAL ACCIDENTS AND INJURIES

An ***industrial accident*** is an unexpected and unintentional event that disrupts work procedures and has the potential to cause damage to property and injury or death to workers. To be classified as an accident, the event does not have to actually cause damage, injury, or death; it only has to have the potential for these consequences. The objective of an occupational safety program is to reduce the incidence of accidents by reducing the hazards that precipitate them. A ***hazard*** is a condition or set of conditions that has the potential for causing an accident or other harmful outcome. The ***danger*** posed by a hazard is the relative exposure or liability to injury, death, and/or damage from that hazard. For example, a railroad crossing is a hazard, but it poses little danger so long as the crossing signal works and drivers pay attention to the signal.

Industrial accidents can cause injuries that are fatal or nonfatal. The Bureau of Labor Statistics maintains statistical records on occupational injuries, illnesses, and fatalities. The rates vary according to industry. A representative listing of fatal and nonfatal injury rates in the United States is compiled in Table 1. In Section 2, we provide a list of industrial illnesses. It should be noted that the statistics in these tables include only private industries (only private industries are covered by OSHA), and so it excludes government occupations, some of which are inherently dangerous (e.g., police, firefighters, military).

The causes of fatal and nonfatal injuries can be identified and classified, and the Bureau of Labor Statistics compiles statistics on the causes. Table 2 lists the major causes. Ranking at the top of the list are transportation incidents (e.g., highway accidents causing death or injury) that occurred during employment. So much industrial commerce involves transportation. Also note that homicides and suicides are high on the list. These are fatal injuries that occurred during work.

TABLE 1 Private Industry Categories with Frequencies of Fatal and Nonfatal Injuries

Industries in Order of Frequency of Fatal Injuries	Fatal Injuries per 100,000 Workers	Industries in Order of Frequency of Nonfatal Injuries	Nonfatal Injuries per 100 Workers
Mining	23.5	Construction	6.9
Agriculture	22.7	Manufacturing	6.4
Construction	12.2	Agriculture	6.0
Transportation	11.3	Transportation	5.8
Manufacturing	3.1	Wholesale, retail trade	5.1
Wholesale, retail trade	2.4	Services	4.3
Services	1.7	Mining	3.8
Finance, insurance, real estate	1.0	Finance, insurance, real estate	1.5

Source: Bureau of Labor Statistics (www.bls.gov), based on 2002 data.

TABLE 2 Causes and Exposures of Fatal and Nonfatal Injuries

Transportation incidents (e.g., collisions)
Overexertion (e.g., back problems from lifting)
Assaults and violent acts (includes homicides and suicides)
Contact with objects and equipment (e.g., struck by objects, compression)
Falls
Exposure to harmful substances and environments (electric shock, temperature extremes, hazardous substances)
Fires and explosions

Why do industrial accidents occur? Three factors are commonly identified in the ergonomics literature as the primary reasons:

- *Human errors.* These factors relate to the worker or workers who are responsible for the operation and make mistakes that are sometimes the direct cause of an industrial accident.
- *Job factors.* This category refers to the kinds of tasks, methods, materials, equipment, and so on that are used in the operation. Some jobs are more dangerous than others.
- *Environmental conditions.* The environment includes lighting, noise, temperature, and other conditions that surround the operation. For example, the dim lighting in a plant made it difficult for a worker to see the obstacle in his path, and he tripped and broke his ankle.

These factors often act in combination, so that the cause of an accident cannot be isolated to a single reason.

In addition to these primary factors, there are organizational variables and management policies that influence the likelihood that an accident will occur in a given company. For example, if management shows little regard for safety in the plant, the company's workforce may develop a culture that disregards hazards, thus increasing the probability of accidents. A model that depicts all of these factors is shown in Figure 1.

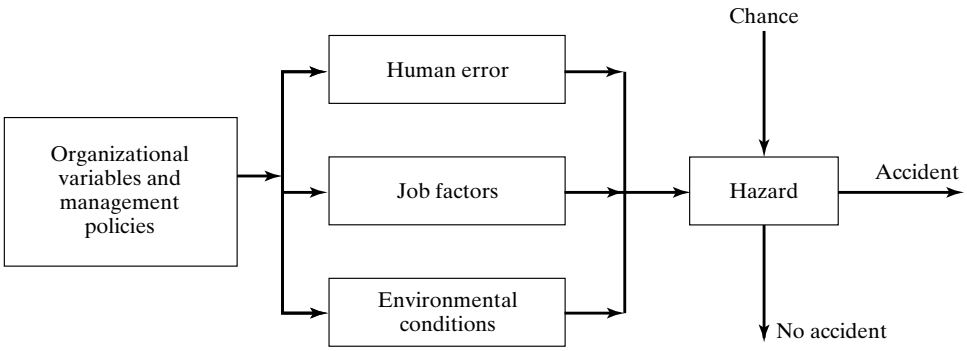


Figure 1 A model depicting the possible factors that contribute to industrial accidents.

1.1 Human Error and Accidents

Human error is often identified as the cause of an industrial accident. In fact, human error probably plays a role in the various other factors identified in Figure 1 that ultimately contribute to an industrial accident. In the preceding example on environmental conditions, the plant engineer who neglected to bring the lighting in the plant up to an acceptable level caused the conditions that resulted in the worker breaking his ankle. Based on this kind of reasoning, it could be argued that all accidents are caused by human error, either directly or indirectly.

In the context of industrial accidents, ***human error*** can be defined as an improper and/or inadvertent human act or decision that has the effect or the potential to reduce effectiveness or safety in the workplace. Human errors can be classified as (1) errors of omission and (2) errors of commission. ***Errors of omission*** occur when a worker fails to take some action that is called for. For example, a home improvement contractor forgets to turn off the circuit breaker to the section of the house that he is rewiring and experiences an electrical shock (defined in Section 1.2). ***Errors of commission*** occur when a worker takes an action that is incorrect. For example, the same home improvement contractor remembers to turn off the circuit breaker, but he switches off the wrong circuit, causing him to suffer an electrical shock.

Among the factors that correlate with accident rates are (1) age, (2) time on the job, (3) fatigue, and (4) stress [9]. Age is a significant factor. Young persons have higher accident rates according to research studies. One can speculate that as people become more mature, they become more cautious and conservative, and their propensity for risk-taking decreases. The net effect is a reduction in accident rates. Contrasting with this trend, however, is that accident rates increase for the elderly, at least in work that requires a high level of physical and cognitive capacity.

Time on the job also correlates with accident rate. A high proportion of industrial accidents occur within the first three years of employment on a new job, with the rate peaking at two to three months [9]. This is explained by the fact that the new worker has completed whatever training program was provided but is still lacking in the experience needed to adequately recognize hazards and select optimal responses.

Finally, fatigue and stress are shorter-term factors that adversely affect accident rates. Fatigue reduces awareness, and stress reduces available attention resources for the task at hand. Both effects increase the likelihood of accidents. The reasons for stress need not be work-related. A worker's personal problems (e.g., financial, marital) may intrude on his or her capacity for safe job performance.

1.2 Job Factors

Accident rates are higher in certain industries because the jobs in those industries are inherently more dangerous. Construction work exposes employees to greater dangers than real estate work. Job factors include the characteristics of the task or job, such as the methods, equipment, and materials used.

Methods. The methods used in a job may be manual, mechanized, or automated, or there may be combinations of these categories for certain tasks. If completely manual, the job may require a high physical workload that leads to physical fatigue, a result that increases the likelihood of human error and accidents. Material handling represents a significant proportion of manual labor tasks in industry (e.g., moving cartons in warehouses, loading and unloading production machines, handling materials at a construction site), and it accounts for a large fraction of occupational accidents and injuries. Common injuries in material handling include (1) musculoskeletal strains and sprains resulting from lifting loads that are too heavy, (2) fractures and bruises caused by loads being dropped and similar mishaps, and (3) lacerations and bruises resulting from efforts to dislodge items in storage and cutting of packaging materials used for items in storage [2].

Mechanized and automated methods can often be used to mitigate the accident and injury rates due to manual material-handling tasks. These methods include the use of powered forklift trucks, conveyors, and cranes. However, material handling by machine carries its own risks. Forklift trucks are driven by humans, and humans commit errors. Aside from the bad driver issues, there are problems with loads not being centered on the forks, overloading, loads stacked too high, and so on. Conveyor systems provide an efficient method to move large amounts of materials in factories, warehouses, and distribution centers. Common causes of conveyor injuries include pinch points between moving and stationary components of the conveyor, loads jamming, loads falling from the conveyor, and workers falling as they try to cross the conveyor. Cranes are used for lifting and moving large loads in a facility, and hazards often arise simply because the loads are heavy. As the vice president of a crane company once said at a material-handling meeting, "We stand behind our product, but we don't stand beneath it."

Equipment. The equipment used by a worker can be the source of hazards in a job. The machine component often plays its role in the worker-machine system through its capacity to apply greater forces and power than a human can apply. The capacity for high forces and power gives rise to equipment hazards. These hazards can be classified into three categories: (1) electrical hazards, (2) mechanical hazards, and (3) temperature hazards.

TABLE 3 Common Effects on Human Body of Various Current Levels for a 1-Second Duration

Current Level (amps)	Effect on Human Body
0.001	Faint tingling sensation.
0.005	Slight shock felt that is disturbing but not painful. Most individuals can let go.
0.006–0.025	Painful shock with loss of muscular control. At the lower end of this range is the "let-go" current level, above which the individual is unable to voluntarily let go of the contact.
0.050–0.150	Extreme pain, respiratory arrest, muscular convulsions, and possible death.
1	Heart fibrillation (lack of rhythmic pumping action), severe muscular convulsions, and death likely.
10	Severe burns, cardiac arrest, and probable death.

Source: Controlling Electrical Hazards, OSHA Publication 3075, U.S. Department of Labor, Washington, DC, 1986.

TABLE 4 Techniques for Reducing Electrical Hazards

Insulate electrical circuits using materials that are suitable for the voltage and conditions (e.g., temperature, humidity, chemicals).
Isolate indoor electrical installations of more than 600 volts so that unqualified personnel cannot gain access (e.g., lock-controlled rooms or screens, elevation above the floor).
Provide proper grounding of equipment to provide a low-resistance path to the earth so that higher than normal voltages will avoid using the human body to complete the circuit to ground.
Use fuses and circuit breakers to limit excessive current flow.
Use appropriate personal protective equipment and clothing.
De-energize electrical equipment before performing maintenance.

Source: Compiled from [2].

Electrical hazards arise because of ***electrical shock***, which is the discharge of electricity through the body that causes a sudden stimulation of the nerves and possible convulsive contraction of the muscles.[2] An electrical shock occurs when the body is part of an electrical circuit, and current passes through the circuit. The severity of the shock usually depends on the level of current, its duration, and the path of the current between the contact points of the body with the circuit. Table 3 describes the common effects on the human body of various current levels for a 1-second duration, where the path of the current is from the hand to the foot. It should be noted that these reactions vary for different individuals. Burns are the most common injury from electrical shock. Electrical burns are caused by (1) current passing through the body tissue, (2) electrical arcs and flashes, (3) contact with hot surfaces of overheated electrical components, and (4) burning clothing caused by electrical contact or arcing. Electrical shock sometimes induces secondary injuries due to involuntary muscle contractions. These muscle contractions can cause falls and impact injuries due to body reactions. Table 4 presents a list of actions that can be taken to reduce or prevent electrical shock hazards.

[2]Much of this section on electrical hazards is based on *Controlling Electrical Hazards*, OSHA Publication 3075, U.S. Department of Labor, Washington, DC, 1986.

Mechanical hazards arise due to the mechanical motions and forces applied by equipment components. Equipment producing these motions and forces include rotating machinery (e.g., lathes, motors, turbines), shearing equipment (e.g., power shears, brake presses), presses (e.g., press forges, stamping presses, injection molding machines), and impact equipment (e.g., stamping presses, forge hammers). The power generated by these kinds of equipment creates the conditions under which accidents can occur. If we add the potential for human error to the mix, there is an increased risk of fatalities and injuries such as scrapes, cuts, lacerations, broken bones, severed limbs, crushed body parts, and eye loss. Table 5 presents various means by which mechanical hazards are reduced.

The category of temperature hazards includes equipment that is too hot or too cold, but it does not include fire hazards, which are generally associated with the materials used in the job rather than equipment. However, it should be noted that fires are sometimes initiated by electrical problems in equipment, resulting in so-called "electrical fires." Equipment that normally operates at elevated temperatures includes processes such as arc welding, oxyfuel welding, brazing, soldering, heat treatment, flame cutting, metal casting, plastic molding, and glassworking. Burns are the obvious types of injuries that occur with equipment that operates hot. Cryogenic equipment operates at temperatures that are well below freezing, and human physical contact with the cold surfaces or materials (e.g., liquid nitrogen) used in the processes causes skin damage.

Materials. Many industries use materials in their normal operations that are classified as hazardous—that present health or safety hazards to workers, the facility in which they are used, or the environment. Materials that cause such hazards to humans can be classified into three categories: (1) corrosive materials, (2) toxic or irritant materials, and (3) flammable materials [5].

Corrosive materials are usually acidic or caustic substances that can burn or damage human tissue. Exposure can occur due to skin contact or inhalation.

Toxic or irritant materials are poisons that disrupt the normal body processes. They include liquids, gases, and solids. ***Toxicology*** is the science that studies poisons and their effects and problems. There are more than 50,000 known toxic substances. Their effects depend on the concentration of the toxic chemical, and in some instances the effects do not manifest themselves for years. Effects may include (1) cancerous tumors and other tumors, (2) embryo damage, (3) irritation of the skin, eyes, or respiratory tract, (4) reduction in mental alertness, (5) altered behavior, (6) general decline in health, (7) reduction in sexual function, and (8) other short-term or long-term illnesses [6].

TABLE 5 Techniques and Devices for Reducing Mechanical Hazards

Design a machine so that its dangerous working parts are not accessible to operators.
Surround the entire machine with enclosures that isolate it from workers.
Use machine guards to enclose certain operations if enclosures for the entire machine are not feasible.
Install interlocks for doors that open to the operation, so that only when the doors are closed can the operating cycle proceed.
Place emergency stop buttons within easy reach of the operator.
Use two-hand start buttons that require both hands to begin the machine cycle. This ensures that both hands are out of the operation area of the machine.

Source: Compiled from [2].

There are three ways in which toxic substances can enter the body: (1) inhalation, (2) absorption, and (3) ingestion. Inhalation is the most rapid entry mode, and since the respiratory and circulatory systems are so closely associated in the lungs, inhaled toxic materials are readily transported throughout the body. Absorption through the skin is less common than inhalation because the epidermis resists absorption, especially through the palms of the hands, where the skin is thicker. Finally, ingestion of toxic materials is also less common because humans tend to exercise reasonable judgments about the things they eat and drink.

Flammable materials are those that present hazards of fire or explosion. Three ingredients are needed to ignite a flammable material: (1) an oxidizer, (2) heat, and (3) a chain chemical reaction. Preventing ignition of flammable materials means denying them of these conditions. There are several ways to remove these conditions: (1) separating the flammable material from the oxidizer, (2) quenching, and/or (3) inhibiting the chain chemical reaction. Table 6 describes the four classes of fires, based on the fuel involved. Different firefighting approaches are required for the each class.

1.3 Environmental Conditions

Environmental conditions are those factors in the immediate surroundings of an operation. They include (1) physical factors and (2) social or psychological factors. Table 7 organizes these environmental factors according to this distinction. If environmental factors are found to adversely affect workers, they will play a role in increasing the risk of industrial accidents.

TABLE 6 Four Classes of Fires

Class	Description
Class A	Burning of solid materials such as wood, coal, and paper.
Class B	Burning of gases or liquids that require vaporization for combustion.
Class C	Class A or B fires involving electrical equipment.
Class D	Burning of easily oxidized metals such as magnesium, titanium, or aluminum.

TABLE 7 Environmental Factors that May Affect the Likelihood of Industrial Accidents

Physical Environmental Factors	Social/Psychological Environmental Factors
Lighting	Management policies and practices
Noise	Social and cultural norms
Temperature and humidity	Morale
Vibration	Training and instruction
Airborne pollutants	Incentives
Fire hazards	Safety programs in place
Radiation hazards	
Conditions that increase the likelihood of falls	

Source: Based on a table in [9].

TABLE 8 Some Common Cumulative Trauma and Other Occupational Disorders

Disorder	Description
Carpal tunnel syndrome	Inflammation of the tendons in the carpal tunnel (in the wrist) resulting in compression of the median nerve, causing pain, swelling, and numbness.
Bursitis	Inflammation of the bursa, which are pouches containing fluid that reduce friction in the joints.
Tendonitis	Inflammation of certain tendons resulting from excessive use.
Trigger finger	Condition in which the person can curl the finger but is unable to straighten it out, except passively.
Neuritis	Inflammation of the nerve in a body member such as the hand.
Lower-back disorders	Condition of the lower back muscles and skeleton causing pain and accounting for a high proportion of all nonfatal occupational injuries. It is estimated that the jobs of as many as 30% of workers in the United States require tasks that risk lower-back disorders.

Source: Compiled from [6] and [8].

2 OCCUPATIONAL DISORDERS AND DISEASES

The injuries and fatalities discussed in the previous section are one-time events or ***single-incident traumas***. Sometimes, the injury or health problem is one that results from a cumulative condition. These types of disorders, which may take years before their effects are evident, are called ***cumulative trauma disorders*** (CTDs). They are also known as repetitive-motion disorders because they are caused by repeated use of certain tendons and nerves, such as those in the fingers, wrist, forearm, upper arm, and shoulder. Occupations in which these body members are used repeatedly and in which CTDs are common include manual assembly with short cycle times, garment sewing, packaging, wiring, and carpentry [6]. In these kinds of manual operations, there are several causal factors leading to cumulative trauma disorders: (1) highly repetitive activity, (2) unnatural joint postures, and (3) application of forces through hinge joints. A list of cumulative trauma disorders is compiled in Table 8.

Occupational diseases usually result from exposure to a substance over an extended period of time. In some cases, the onset of the disease is delayed for many years. Some of these maladies are discussed in the context of toxic substances in Section 1.2. The more common occupational diseases are identified and briefly described in Table 9.

3 OCCUPATIONAL SAFETY AND HEALTH LAWS AND AGENCIES

This section focuses on the important legislation that has been enacted to promote safer and more healthful conditions in the workplace. In most cases, the laws have created governmental agencies to promulgate and enforce the policies and standards established by the legislation. Before these laws and agencies came into being, it is generally acknowledged that working conditions were neither safe nor healthful in many industries throughout the world. The two most important pieces of legislation enacted into law in the United States are (1) the various state workers' compensation laws and (2) the Occupational Safety and Health Act (OSHAct), a federal law.